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Mayank Yadav

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Education

Delhi Technological University May 2027
B. Tech. Computer Science and Engineering *CGPA: 8.99*

Central Board of Secondary Education (Class 12th) May 2023
Computer Science, Mathematics, Physics, Chemistry, English, Physical Education *Grade: 97.4%*

Central Board of Secondary Education (Class 10th) August 2021
English, Hindi, Mathematics, Science, Social Science, Sanskrit *Grade: 97.8%*

Technical Skills

Languages: C/C++, Python, Java, HTML, Tailwind-CSS, JavaScript, TypeScript, SQL, LaTeX
Tools: Git/GitHub, Docker, Kubernetes, Harness CI/CD, Google Cloud Platform, Linux, VS Code
Frameworks: React, Node.js, Flask, FastAPI
Libraries: PyTorch, Detectron2, TensorFlow, Keras, pandas, NumPy, Matplotlib, NLTK, Scikit-Learn, OpenCV, LangGraph

Experience

Open Source Developer, Summer of Bitcoin (SeedSigner) *May 2026 – August 2026*

- Developed a stateless secure bootloader for *ESP32-P4 (RISC-V)* and *ESP32-S3 (Xtensa)* to verify and execute MicroPython directly from SD into PSRAM via Cache MMU remapping, preventing secret persistence in flash.
- Resolved critical FreeRTOS early-boot deadlocks and cache coherency faults by safely intercepting ESP-IDF hardware resets and manually aligning `.data` and `.bss` memory segments during the MMU handoff.
- Devised a physical anti-phishing defense that fills unallocated flash with TRNG data to compute a 4-word BIP-39 visual proof on boot, validating tamper detection against 16 hardware attack vectors (TOCTOU, bit-flips, evil-maid swaps).

Technology Program Intern, Wells Fargo, Hyderabad *May 2026 – July 2026*

- Built a full-stack agentic intake workflow using React, FastAPI, and LangGraph to automate L3 data application triage, cutting initial Jira screening from 10+ days to near-instant validation.
- Designed an 11-stage autonomous pipeline featuring an upfront validation layer to automate the end-to-end request lifecycle across mapping, design, QA, and peer review checkpoints.
- Automated generation of Data Warehouse Design workbooks and QA SQL packs by integrating with an existing Informatica generator to produce downstream ETL artifacts.
- Orchestrated the application's release on Kubernetes using GitHub Actions and Harness CI/CD, and added a role-based admin dashboard with Server-Sent Events (SSE) for real-time tracking.

Web Developer Intern, DTU, Delhi *October 2025 – March 2026*

- Co-Developed a centralized IQAC data management suite using the MERN stack, deployed across 15+ departments and used by 500+ faculty members for structured academic and administrative data collection with aggregated dashboards.
- Engineered a 3-tier APAR workflow system (Faculty → Reporting Officer → Reviewing Officer) with automated data pre-filling from the IQAC DB, ensuring data consistency and reducing manual redundancy.
- Implemented JWT-based authentication, role-based access control (RBAC), and WebSockets (Socket.io) to enable secure access control and real-time update propagation across hierarchical user roles.

Research Intern, MLR, DTU, Delhi *June 2025 – November 2025*

- Developed 5+ deep learning models for vision-based pose estimation of non-cooperative satellites using PyTorch framework.
- Implemented and optimised multiple SOTA deep learning architectures including SPN, PVNet, ViT Pose, and SWIN Transformer for robust 6-DoF satellite pose estimation; applied RANSAC and Perspective-n-Point solver.
- Designed a novel SWIN-based pixel-voting mechanism that integrates transformer attention with 4-layer decoder for PVNet's voting strategy.
- Conducted experiments on SPEED dataset containing synthetic spacecraft images and SPEED+ dataset with hardware-in-the-loop test images.

Projects

Sherlock | GitHub Link | Deployed Link | *TypeScript, Next.js, React, Tailwind CSS, Vitest* March 2026

- Built a local-first Bitcoin chain analysis tool that parses raw Bitcoin Core `.dat` files (blk, rev, xor) to reconstruct transaction graphs without full node indexing or external API dependencies.
- Implemented 10+ rule-based on-chain heuristics including CoinJoin fingerprinting, UTXO consolidation, peeling chain detection, batch payments, and OP_RETURN extraction for transaction pattern classification.
- Engineered an interactive Next.js 15 dashboard with TypeScript and Tailwind CSS v4 for visualizing block data, transaction flows, and script distributions, with Vitest test coverage and deployed on Vercel.

GraphGrad | GitHub Link | Deployed Link | *TypeScript, Next.js, React, Tailwind CSS, React Flow, Vitest* March 2026

- Built an interactive computation graph visualizer for learning forward passes and backpropagation, parsing mathematical equations into React Flow graphs with real-time gradient flow visualization.
- Implemented a custom autodiff engine supporting 13 operations with forward evaluation and backward gradient propagation, inspired by Karpathy's micrograd.
- Developed with Next.js 16, React 19, TypeScript, and Tailwind CSS v4, featuring light/dark mode, equation-to-graph generation via mathjs, and a Vitest test suite.

Vulcan-16 | GitHub Link | Deployed Link | *HDL, Python, Assembly, Compiler Designing, Jack, Git, React, Flask* January 2025

- Architected a 16-bit Harvard-architecture CPU in HDL, including core logic, memory, and I/O interfaces.
- Created a 3-phase compilation pipeline and integrated web-based development environment.
- Designed an OS with 8 modular services for memory management, graphics, I/O, and utilities.

Mentoring Portal | GitHub Link | Deployed Link | *React, Tailwind, Node.js, Redux Toolkit, Express, MongoDB* December 2024

- Developed a full-stack mentorship portal enabling secure mentor-student engagement with Q&A forums and resource library for 5 competitive exams.
- Implemented end-to-end features using React, Redux Toolkit, Express, and MongoDB, delivering 4+ core modules.
- Deployed and maintained a scalable platform serving 50+ underprivileged students for career guidance.

Int-O-View | GitHub Link | Deployed Link | *Python, TypeScript, React, Express, Flask, MongoDB, LangChain* August 2024

- Engineered an AI-Agent based interviewer app using Gemma 2, Qwen QwQ, and ElevenLabs for realistic voice synthesis.
- Integrated Supabase embeddings and LangGraph StateGraph for context retrieval powering 100 real-time interview simulations and LLM-driven question generation.
- Collaborated in a 5-member team to secure 1st place in the SIH internal round, surpassing 200+ competing teams.

Lung Cancer Detection | GitHub Link | *Python, PyTorch, Detectron2, Pydicom* March 2025

- Transformed the LIDC-IDRI dataset by converting 3D DICOM images into 2D JPEG format using advanced windowing techniques, and reformatted XML annotations into the COCO JSON standard.
- Developed a two-stage lung nodule detection pipeline having a Cascaded Mask R-CNN (trained on VESSEL12 dataset) for lung segmentation, and a Faster R-CNN for nodules localisation and prediction of 5 key radiological features—malignancy, margin, sphericity, texture, and spiculation—to aid in diagnostic decision-making.
- Achieved segmentation accuracy of 81.98% and bounding box accuracy of 93.62% on the test set in the segmentation model, and Validation set bounding box AP50 scores of 24.97 (malignancy), 24.34 (sphericity), 23.16 (margin), 23.26 (spiculation), and 16.76 (texture).

Summer ML Projects | Github Link | *Python, PyTorch, Transformers* June 2024

- Developed a collection of 7 machine learning projects during the Summer School on AI by MLR, DTU, demonstrating core concepts in ML and deep learning.
- Implemented a Generative Pre-trained Transformer 2 (GPT-2) from scratch for text generation and a Vision Transformer (ViT) for image classification.
- Built 3 different image captioning models implementing "Where to put the Image in an Image Caption Generator" paper using a combination of CNN and RNN/LSTM architectures and trained on Flickr 8k images.

Facial Expression Recogniser | GitHub Link | Kaggle Link | *Python, TensorFlow, Keras, OpenCV, SMOTE* May 2024

- Built a real-time facial emotion recognition system on the FER2013 dataset, addressing severe class imbalance using SMOTE, random oversampling, and data augmentation to improve model generalization.
- Developed and evaluated 50+ distinct neural network architectures from scratch, spanning fully connected networks to convolutional networks, systematically comparing depth, layer size, and activation configurations.
- Explored transfer learning with pre-trained Inception and VGG16 models, achieving competitive accuracy through fine-tuning, with a public Kaggle notebook documenting the full experimentation pipeline.

Open Source Contributions

Shopstr | *TypeScript, Next.js* April 2026

- Contributed 7+ merged bug-fix and feature PRs to a global permissionless Nostr marketplace for Bitcoin commerce (100+ stars), covering server-side pagination with advanced database filtering, profile image upload, and marketplace listing routing.
- Fixed client-side issues including navbar profile flickering and deduplication of repeated profile-fetch requests, improving UI stability and load performance.

BitcoinFuzz | *C++, Python, Docker* May 2026

- Merged 2 PRs to a differential fuzzing project (84+ stars) that compares Bitcoin/Lightning protocol implementations, fixing the Docker build by updating `auto_build.py` to initialize the secp256k1 submodule for custom-mutator targets and correctly skipping empty-input/empty-output PSBT matches during fuzzing.

SeedSigner Emulator | *Python* April 2026

- Fixed the MockST7789 initialization `TypeError` by correcting the display library's constructor call, resolving a blocking startup failure in the emulator.

NN-SVG | GitHub PR Link | Deployed Link | *JavaScript, Three.js* July 2025

- Delivered a key feature to NN-SVG (5.6k+ GitHub stars), a tool used by ML researchers to generate publication-ready neural network diagrams, enabling users to upload and render their own images as the input layer texture.

Achievements

NTSE Scholar: Recognized for outstanding performance in the NCERT's national-level scholarship examination, placing among the top 1000 scholars.

Reliance Foundation Undergraduate Scholarship: Selected as one of 5,000 nationwide for academic support and leadership development.

CBSE Certificate of Merit in Mathematics: Achieved 100 in Mathematics, ranking in the top 0.1 % in the 2023 All India Secondary School Examination.

Position of Responsibility

AlgoRave | *Co-Head* September 2023 - Present

- Authored and validated problems for 3+ contests on CodeForces, utilizing Polygon platform for problem development and testing.
- Drafted 2+ detailed editorials for the contest problems, including solution approaches, algorithms, and code implementations.
- Organised contests that attracted 100+ participants, enhancing engagement and competitive programming culture within the university

Licenses & Certifications

Machine Learning Specialization | *Stanford Online / DeepLearning.AI* *Dec 2023*

- Supervised Machine Learning, Advanced Learning Algorithms, Unsupervised Learning & Recommenders

Foundations of Cybersecurity | *Coursera / Google* *Oct 2025*
Hugging Face Agent Course | *Hugging Face* *May 2025*
Fundamentals of MCP | *Hugging Face* *May 2025*
Introduction to Deep Learning | *Kaggle* *Jan 2024*
Getting Started with AI on Jetson Nano | *NVIDIA* *Apr 2025*
Google Cloud Computing Foundations (5 courses), Prompt Engineering & GenAI | *Google* *Oct 2023*
The Joy of Computing using Python | *NPTEL* *Nov 2025*
Human Computer Interaction | *NPTEL* *May 2026*